

§8 jagged — per-step GPU-time in absolute ms, in run order (GB300 16-GPU)

bar height = step wall-clock · the compute bands (HSTU/GEMM/ELEM) are steady; step-to-step variation is in the idle + NCCL(exposed) bands (host gaps / comms stalls)

IDLE NCCL(exp) HSTU GEMM ELEM EMB OTH

